SIMATS ENGINEERING

ASSESSMENT TOOL 1 - ANSWERS

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07
COURSE OUTCOME COVERED: CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)
Assessment Tool Weightage: 40%

ANALYTICAL PROBLEM SOLVING - ANSWER KEY

1. University network 192.168.20.0/24 — Lab A (/26), Lab B (/27), Lab C (/28)

Answer:

- Lab A (/26): Network address 192.168.20.0, Broadcast address 192.168.20.63
  - Valid host range: 192.168.20.1 – 192.168.20.62
  - Usable hosts: 62
- Lab B (/27): Network address 192.168.20.64, Broadcast address 192.168.20.95
  - Valid host range: 192.168.20.65 – 192.168.20.94
  - Usable hosts: 30
- Lab C (/28): Network address 192.168.20.96, Broadcast address 192.168.20.111
  - Valid host range: 192.168.20.97 – 192.168.20.110
  - Usable hosts: 14

2. System 1 (192.168.1.130/26) and System 2 (192.168.1.190/26)

Answer:

Subnet mask 255.255.255.192 = /26 → block size 64. The /26 blocks in this octet are 0–63, 64–127, 128–191, 192–255.

- System 1 (192.168.1.130) falls in the 128–191 block:
  - Network address: 192.168.1.128
  - Broadcast address: 192.168.1.191
  - Valid host range: 192.168.1.129 – 192.168.1.190
  - 192.168.1.130 lies within this range → VALID
- System 2 (192.168.1.190) falls in the same 128–191 block:
  - Network address: 192.168.1.128
  - Broadcast address: 192.168.1.191
  - Valid host range: 192.168.1.129 – 192.168.1.190
  - 192.168.1.190 lies within this range → VALID (it is the last usable host)
- Both systems belong to the SAME subnet: 192.168.1.128/26
  - Usable host addresses in the subnet: 62

3. Organization block 172.16.0.0/16 — HR (/25), Sales (/26), IT (/27), Admin (/28)

Answer:

- HR (/25): Network 172.16.0.0, Broadcast 172.16.0.127
    - Valid host range: 172.16.0.1 – 172.16.0.126 | Usable hosts: 126
- Sales (/26): Network 172.16.0.128, Broadcast 172.16.0.191
    - Valid host range: 172.16.0.129 – 172.16.0.190 | Usable hosts: 62
- IT (/27): Network 172.16.0.192, Broadcast 172.16.0.223
    - Valid host range: 172.16.0.193 – 172.16.0.222 | Usable hosts: 30
- Admin (/28): Network 172.16.0.224, Broadcast 172.16.0.239
    - Valid host range: 172.16.0.225 – 172.16.0.238 | Usable hosts: 14
- Remaining unused IP address range: 172.16.0.240 – 172.16.255.255

4. 172.16.10.0/24 subnetted with /27

Answer:

- Total number of subnets created: 2^3 = 8 subnets (borrowed 3 bits from /24 to /27)
- Usable hosts per subnet: 2^5 − 2 = 30
- The 8 subnets are: 172.16.10.0, .32, .64, .96, .128, .160, .192, .224 (each /27)
- IP address 172.16.10.78 belongs to the subnet 172.16.10.64/27 (range 64–95):
    - Network address: 172.16.10.64
    - Broadcast address: 172.16.10.95
- Is 172.16.10.95 in the same subnet range?
    - Yes — 172.16.10.95 lies within the 172.16.10.64/27 block; however it is the BROADCAST address of that subnet, not a usable host address.

5. IPv6 address 2001:0db8:0000:0000:0000:ff00:0042:8329

Answer:

- Compressed notation: 2001:db8::ff00:42:8329
- Expanded (full) notation: 2001:0db8:0000:0000:0000:ff00:0042:8329
- Network prefix (/64): 2001:0db8:0000:0000::/64 (i.e., 2001:db8::/64)
- Total possible host addresses within this /64 network: 2^64 = 18,446,744,073,709,551,616 host addresses